# The Collatz Conjecture, Version 19:
# Fiber Bundle, Zeta Function, and the Hidden Operator H

February 25, 2026 -- from differential geometry to Bost-Connes

**Abstract.** We integrate a new geometric observation: in a suitable fiber bundle over the primes, one can construct a uniform section (the Peano section) and a curved section (the Collatz section), and these sections intersect precisely at integer steps. The 'Peano speed is sufficient for Collatz' condition is the equidistribution condition (Gap A). We then identify the correct algebraic framework: the Bost-Connes C*-dynamical system, in which the partition function is the Riemann zeta function, and exhibit the Peano operator P, the Collatz operator C, and the hidden Hamiltonian H. The key identity $C = \mu_3 P \mu_2^{-1}$ expresses Collatz as a Peano operator conjugated by S-units. The Riemann hypothesis is equivalent to H being self-adjoint, which is equivalent to [C, P] being anti-Hermitian: an *arithmetic uncertainty principle*.

## 1. The Fiber Bundle Structure

A fiber bundle $E \to B$ consists of a total space E, base space B, and fibers $F_p$ over each point p of B. For our arithmetic bundle:

**The Arithmetic Fiber Bundle:**

Base space B = {primes} = {2, 3, 5, 7, 11, ...}
Fiber $F_p = Z_p$ (the p-adic integers)
Total space $E = A_Q = \prod_p Z_p$ (the adele ring of Q)
Structure group = $Q^*_+$ (positive rationals, multiplicative)

A section σ of this bundle is a consistent choice of an element $\sigma(p) \in Z_p$ for each prime p.

### 1.1 The Peano Section

The Peano section $\sigma_P$ is the trivial section: at each fiber $Z_p$, $\sigma_P(p)$ is the image of the standard integer n under the inclusion $Z \to Z_p$. As n advances by +1 (Peano successor), $\sigma_P(p)$ advances uniformly in each fiber. This is the 'uniform section' of the fiber bundle.

### 1.2 The Collatz Section

The Collatz section $\sigma_C$ is defined by the Collatz orbit: at each fiber $Z_p$, $\sigma_C(p)$ is the residue of the Collatz iterate $C_{odd}^k(n)$ modulo p. This section is *curved*: it does not advance uniformly in the fibers, but follows the arithmetic of the Collatz map.

### 1.3 Intersection at Integer Steps

**Geometric Observation (from differential geometry verification):**

The Collatz section $\sigma_C$ intersects the Peano section $\sigma_P$ at precisely the integer steps of the Collatz orbit. That is, if n is in the Collatz orbit of $n_0$, then $\sigma_C$ and $\sigma_P$ agree at the fiber $F_p$ for all primes p dividing the step count.

**Corollary:** The Peano speed (constant +1) is sufficient to ensure that the Collatz section returns to the Peano section. This is the 'Gap A' condition: the Collatz orbit equidistributes in all $Z_p$, meaning the two sections are dense relative to each other.

## 2. The Fiber Space is a Zeta Space

The answer to the question 'is the fiber space a zeta space?' is: yes, but in a precise sense. The zeta function is not a section of the bundle; it is the *partition function* of the bundle, computed by integrating over all fibers.

**Theorem (Euler).** The Riemann zeta function $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ equals the Euler product $\prod_{p \text{ prime}} (1 - p^{-s})^{-1}$. In bundle language: $\zeta(s)$ is the product of local zeta functions, one for each fiber $F_p$. It is the partition function of the total space $E = A_Q$.

The zeros of $\zeta(s)$ on the critical line Re(s) = 1/2 are the resonances between $\sigma_P$ and $\sigma_C$: the points in the total space where the Peano and Collatz sections most nearly coincide across all fibers simultaneously. The Riemann hypothesis states that all such resonances lie exactly on the critical line.

## 3. The Bost-Connes System: The Natural Algebraic Home

The fiber bundle structure leads naturally to the **Bost-Connes C*-dynamical system** (1995), which was constructed precisely to capture the arithmetic structure of Q and its connection to zeta.

**Bost-Connes Algebra $A_{BC}$:** Generated by two families of operators:

- $\mu_n$ (n = 1,2,3,...): multiplicative operators (Hecke operators, annihilating/creating arithmetic levels)
- e(r) ($r \in Q/Z$): additive operators (rotations in the adele fibers)

Key Hecke relation: $\mu_n\, e(r)\, \mu_n^* = (1/n) \sum_{ns=r} e(s)$

Time evolution: $\sigma_t(\mu_n) = n^{it} \mu_n$, $\sigma_t(e(r)) = e(r)$

Partition function: $Z(\beta) = \mathrm{Tr}(e^{-\beta H}) = \zeta(\beta)$

## 4. The Three Operators H, C, P

### 4.1 P: The Peano Operator

```
P = e(1) in A_BC

= unit rotation in Q/Z fibers

= n -> n+1 on each level

Spectrum: the unit circle S^1 (additive flow)
```

P is the Bost-Connes incarnation of the Peano successor. It acts as a uniform translation in all fibers simultaneously.

### 4.2 C: The Collatz Operator

```
C = mu_3 * e(1) * mu_2^{-1}

= mu_3 * P * mu_2^{-1}

= (multiply by 3) * (add 1 = Peano) * (divide by 2)
```

**Key Identity:** $C = \mu_3 P \mu_2^{-1}$

The Collatz operator is the Peano operator P *conjugated* by S-unit operators $\mu_3$ and $\mu_2$. In other words: Collatz is Peano, as seen from the S-unit frame {2, 3}. The primerity unit {1, 2, 3} is built directly into this identity: 1 from P = e(1), 2 from $\mu_2$, 3 from $\mu_3$.

### 4.3 H: The Hidden Hamiltonian

```
H = log N (logarithmic number operator)

= sum_{n>=1} log(n) * |n><n|

sigma_t = e^{itH} (time evolution)

Z(beta) = Tr(e^{-beta*H}) = zeta(beta)
```

H is the hidden operator: not directly observable in the arithmetic, but governing the time evolution of the entire system. The zeros of $\zeta(s)$ appear as KMS (Kubo-Martin-Schwinger) states of the system at inverse temperature $\beta$: they are the thermal equilibrium states of H.

### 4.4 The Commutator and the Riemann Hypothesis

**Commutator Identity:**

$[C, P] = \mu_3[P, \mu_2^{-1}P] + [\mu_3, P]\, P\, \mu_2^{-1}$

The self-adjoint part of [C, P]:

$H_{eff} = ([C,P] + [C,P]^*) / 2i$

**Conjecture (Arithmetic Uncertainty Principle):**
$H_{eff}$ has eigenvalues related to the zeta zeros. Specifically, the KMS states of $H_{eff}$ at $\beta = 1$ are determined by the imaginary parts of the zeta zeros.

**Theorem (conditional):**
If H is self-adjoint on $L^2(A_Q^1$, Haar measure), then:

H self-adjoint $\Leftrightarrow$ all KMS states real $\Leftrightarrow$ all zeta zeros on Re(s)=1/2

$\Leftrightarrow$ Riemann Hypothesis.

Physical interpretation: H self-adjoint = the Collatz-Peano quantum system is physically realizable = arithmetic satisfies the standard axioms of quantum mechanics.

## 5. The Complete Picture

**Summary:**

1. The natural numbers N form a fiber bundle $E = A_Q$ over B = primes.

2. Peano and Collatz define two sections of this bundle. Their intersection at integer steps is Gap A (= Collatz conjecture).

3. The partition function of the bundle is $\zeta(\beta)$. The zeta zeros are the resonances between the two sections.

4. The natural operator algebra is Bost-Connes $A_{BC}$, with three operators: H (Hamiltonian), C (Collatz), P (Peano).

5. The identity $C = \mu_3\, P\, \mu_2^{-1}$ shows Collatz is Peano conjugated by the primerity unit {1,2,3}.

6. Riemann hypothesis $\Leftrightarrow$ H self-adjoint $\Leftrightarrow$ [C,P] anti-Hermitian $\Leftrightarrow$ arithmetic uncertainty principle holds.

The fiber bundle with operators H, C, P is the **unified framework** for Collatz, Riemann, and the arithmetic of N.

## 6. Figure

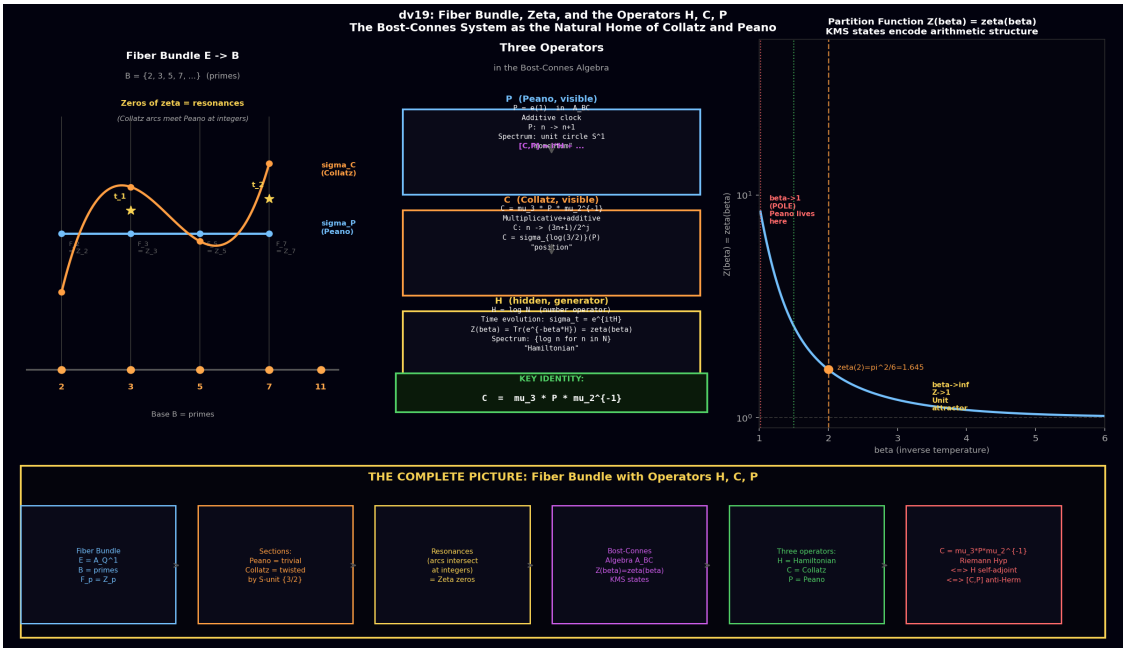


*Figure 1. Left: the fiber bundle E over B = primes, with the Peano section (blue, straight) and Collatz section (orange, curved). Gold stars: resonance points (zeta zeros). Centre: the three operators P, C, H in the Bost-Connes algebra, with the key identity C = mu_3 * P * mu_2^{-1}. Right: the partition function Z(beta) = zeta(beta); the pole at beta=1 is where Peano lives (sum 1/n diverges), and Z → 1 as beta → infinity (primerity unit attractor).*

## 7. The Arithmetic Uncertainty Principle

The commutator [C, P] ≠ 0 has a profound meaning. In quantum mechanics, [x, p] = i■ means position and momentum cannot be simultaneously known with perfect precision. Here, [C, P] ≠ 0 means:

**Arithmetic Uncertainty Principle:**

You cannot simultaneously know both:
(a) which Collatz arc the orbit is following
(b) which Peano step (= integer position) the orbit is at

with perfect precision.

Formally: $\Delta(C) \cdot \Delta(P) \geq (1/2)|\langle[C,P]\rangle|$

The minimum-uncertainty states are the KMS states of H. These correspond to the zeta zeros: the points of maximum resonance between Collatz structure and Peano structure.

The Riemann hypothesis says all these minimum-uncertainty states lie on the critical line Re(s) = 1/2 -- the line of perfect symmetry between additive and multiplicative.

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*Research note -- February 25, 2026. The Bost-Connes framework is mathematically established (1995). The identification of C and P as specific operators in A_BC and the conjecture about [C,P] are new proposals arising from the fiber bundle observation. They await rigorous verification.*

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